Hypothesis Stress-Test & Investment Decision Report v3.0.6.8

Powered by Vraimony
Protocol: ImmuneErrorRadar falsification protocol v3.0.6.9.7 | method=Vraimony methodology | posture=standard_followed_not_copied
Falsification receipt: id=ierfp-d6821ae9d409a436ad2d8b71 | contentSha256=d6821ae9d409a436ad2d8b71 | signedByVraimony=false

Documentation standard: ERF-inspired evidence receipt discipline: hash-first, portable, replayable, claim-bounded.
Citation line: This report follows the ImmuneErrorRadar falsification protocol, powered by Vraimony, based on the Vraimony evidence-receipt methodology.
Vraimony boundary: This receipt documents a falsification workflow and report payload hash; it is not medical advice, not clinical validation, not wet-lab evidence, and not proof of biology.
Report mode: verification-only | evidence-readiness receipt | investment-triage verdict | no discovery claim | no clinical advice

FINAL INVESTMENT DECISION v3.0.6.7: DO_NOT_INVEST_CLAIM_KILLED | claim=claim killed as written for current data | killedClaim=CK2_CSNK2A1_directly_drives_APM_loss_and_worse_PFS_in_this_dataset | criteria=kill:4/invest:1 | next=Do not invest in CK2. If continuing, invest only in mechanism-agnostic APM data, not CK2 experiments. | probabilityReported=false
PREPRINT READINESS v3.0.6.8: PREPRINT_READY_CLAIM_KILLED_AS_WRITTEN | title=Hypothesis Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true | residualDataFollowUp=false | wetLabAllowed=false | probabilityReported=false
Publication data fuel v3.0.6.9: route=CK2_APM_FALLBACK | state=CK2_MECHANISM_REOPEN_GATES_INJECTED_CLAIM_REMAINS_KILLED | syntheticData=false | fakeReplication=false | defense=CK2 mechanism remains killed because external CK2/APM plausibility requires perturbation/protein/IFN context that current static bulk GSE85047 does not execute. | APM multi-cohort fuel v3.0.6.9.6: state=APM_MULTI_COHORT_DATA_FUEL_INJECTED_VALIDATION_TARGETS_ONLY | syntheticData=false | fakeReplication=false | targets=medulloblastoma, Wilms tumor/nephroblastoma, Ewing sarcoma | boundary=data-fuel only, not replication.

Pediatric cross-tumor scope v3.0.6.9.6: state=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE | crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false
Required external pediatric tumors: medulloblastoma,Wilms tumor / nephroblastoma,Ewing sarcoma

Investment triage v3.0.6.6: verdict=KILL_CK2_MECHANISM_DO_NOT_INVEST | killedClaim=CK2_CSNK2A1_directly_drives_APM_loss_and_worse_PFS_in_this_dataset | basis=kill criteria dominate and no defensible next investment exists for the supplied claim; kill=4, invest=1 | next=Do not invest in CK2. If continuing, invest only in mechanism-agnostic APM data, not CK2 experiments. | probabilityReported=false
Question understood: true
Question: In pediatric neuroblastoma GSE85047, high CK2 (CSNK2A1) expression suppresses MHC-I antigen presentation by phosphorylating and destabilizing B2M and HLA-A. Tumors with high CSNK2A1 but low B2M/HLA-A expression show worse progression-free survival independently of MYCN amplification, reflecting an immune-invisible phenotype driven by constitutive CK2 activity

Question understanding v3.0.3.97: CK2_IFN_CONTEXT_APM_REGULATION | confidence=0.95
Mechanistic text understanding v3.0.4.3: C2S_CK2_INTERFERON_APM_HEATING_HYPOTHESIS | confidence=0.95
Text source context: Google DeepMind/Yale C2S-Scale style text | prior=AI_REPORTED_LAB_VALIDATED_CONTEXT_DEPENDENT_APM_UPREGULATION
Text genes understood: CSNK2A1,B2M,HLA-A,IFNGR1,TAP1
Mechanism claim parsed: CK2 inhibition plus an interferon-positive immune context may increase MHC-I/APM expression; phosphorylation/destabilization of B2M/HLA-A is not directly testable here.
Local proxy tests:
CK2_APM_CONTEXT_PROXY:PROXY_ONLY_UPSTREAM_GENE_MISSING_IF_CSNK2A1_NOT_MAPPED | APM_LOW_FALLBACK:FALLBACK_DIRECT_EXPRESSION_SCORE
Not directly testable here: drug synergy: silmitasertib + interferon,phosphorylation/destabilization of B2M/HLA-A,MHC-I protein surface abundance,immune-context-positive perturbation response
Why this route: CK2,CSNK2A1,MHC-I,antigen presentation
Cancer genes understood: B2M,HLA-A,MYCN,CSNK2A1,IFNGR1,TAP1
Dataset/outcome resolved: GSE85047 | PFS | covariates=MYCN
Literature axis basis: C2S/CK2/silmitasertib texts are decomposed into an upstream regulator claim plus an APM expression proxy test.
Understanding caveats:
drug_plus_interferon_synergy_not_testable_with_static_expression_only
Hypothesis tested: CK2_CSNK2A1_IFN_APM_PFS_MYCN_GSE85047
Primary axis: B2M_MHCI_NK_ESCAPE | route=PRIMARY_AXIS_CK2_IFN_CONTEXT_APM_REGULATION_TEXT_PROXY
Genes understood: B2M,HLA-A,MYCN,CSNK2A1,IFNGR1,TAP1
Dataset/model: GSE85047 | outcome=PFS | model=CK2/APM text proxy relaxed screen: APM-low fallback only because CSNK2A1/interferon perturbation is not mapped in selected matrix
Initial result: INITIAL_EXPLORATORY_ADVERSE_SIGNAL_REVIEW_ONLY | priority=SECONDARY_LEAD_REVIEW_ONLY
Triage signal: REVIEW_ONLY_DIRECTIONAL_SCREEN | level=REVIEW_ONLY_DIRECTIONAL_SCREEN_CANDIDATE | verdict=REVIEW_ONLY_COMPUTATIONAL_ROUND_OFFICIAL_CLAIM_BLOCKED
Signal promotion: rawAllowed=true | forKilledClaimAllowed=false | residualDataFollowUpAllowed=false | nextComputationalRoundCandidate=true | wetLabCandidate=false | officialClaimAllowed=false
Directional decision v3.0.4.1: APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | local=PRIORITIZE_APM_SIGNAL_BUT_DO_NOT_ATTRIBUTE_TO_MECHANISM | rank=3
Best direction scan: APM_COMPOSITE_PFS_MYCN_GSE85047 | STRONG_EXPLORATORY_DIRECTION | score=12.3647
Directional ranking table: #1 APM composite:STRONG_EXPLORATORY_DIRECTION | #2 B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION | #3 CK2/APM fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #4 HDAC/APM fallback:APM_FALLBACK_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED | #5 NK effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE | #6 γδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED | #7 CD276/B7-H3:SUBGROUP_ONLY_DIRECTION
Lead label: APM_FALLBACK_STRONG_SIGNAL_MECHANISM_NOT_DIRECTLY_TESTED
Directional why: Strong APM fallback signal is present, but it must not be attributed to the upstream mechanism without mechanism-specific probes or assay evidence:

delta=0.191176, rawHR=1.4159/p=0.0124, adjustedHR=1.1971/p=0.246.
Directional next action: Keep APM as the executed lead; acquire mechanism-specific probes/assays before attributing signal to CK2/HDAC or other upstream mechanisms.
Proxy attribution v3.0.4.4: FALLBACK_APM_PROXY_EXECUTED | signalSource=APM_COMPOSITE_FALLBACK | mechanismGeneTested=false | label=APM_FALLBACK_STRONG_SIGNAL_CK2_MECHANISM_NOT_DIRECTLY_TESTED
Probe lookup v3.0.5.0: resolvedVector=8 | resolvedButVectorMissing=20 | probeUnknown=0 | absentOrReporterUnavailable=2 | staticManifestLocalResolve=TARGETED_STATIC_MANIFEST_ATTACHED | runtimeNetwork=false | legacyAutoFetchFlag=true
Probe vector binding v3.0.5.1: columnIndexPresent=true | resolvedProbeIds=28 | boundVectors=8 | resolvedButColumnMissing=20 | aliasBoundVectors=0 | nextBlocker=PARTIAL_BINDING_REMAINING_GENES_NEED_COLUMN_EXTRACTION
Evidence fuel v3.0.6.0: CK2_APM_FALLBACK | APM_FALLBACK_FUEL_AVAILABLE_CK2_MECHANISM_NOT_BOUND | expected=APM_FALLBACK_SIGNAL_MECHANISM_NOT_ATTRIBUTABLE | boundary=DATA_FUEL_IS_TEST_RECIPE_AND_SELECTED_RUNTIME_COVERAGE_NOT_TIER2_PROOF
Executed local directions v3.0.4.4: #1 APM composite:STRONG_EXPLORATORY_DIRECTION[APM_COMPOSITE_DIRECT_PROXY] | #2 B2M/TAP1/NK:STRONG_EXPLORATORY_DIRECTION[B2M_TAP1_NK_DIRECT_PROXY] | #5 NK effector:DIRECTIONAL_LEAD_CONFOUNDER_SENSITIVE[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROX | #6 γδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED[GAMMA_DELTA_S | #7 CD276/B7-H3:SUBGROUP_ONLY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY] | #8 ODC1/polyamine:WEAK_EXPLORATORY_DIRECTION[DIRECT_OR_ROUTE_SPECIFIC_SELECTED_MATRIX_PROXY]
Mechanistic hypotheses needing assets v3.0.4.4: CK2/APM fallback:APM_FALLBACK_STRONG_SIGNAL_CK2_MECHANISM_NOT_DIRECTLY_TESTED | HDAC/APM fallback:APM_FALLBACK_STRONG_SIGNAL_EPIGENETIC_MECHANISM_NOT_DIRECTLY_TESTED | γδ/NKG2D:GAMMA_DELTA_SURROGATE_DIRECTION_TRGC_TRDV_FULL_MAPPING_REQUIRED | NECTIN2/TIGIT:NECTIN2_TIGIT_NOT_TESTABLE_PROBE_AND_SINGLE_CELL_CONTEXT_REQUIRED
N=272 | events=96 | highEventRate=0.448529 | lowEventRate=0.257353 | delta=0.191176
Cox first-pass: state=RELAXED_COX_FIRST_PASS_EXECUTED_REVIEW_ONLY | HR=1.4159 | p=0.0124 | rows=272 | events=96
MYCN-adjusted Cox first-pass: state=RELAXED_COX_FIRST_PASS_EXECUTED_REVIEW_ONLY | HR=1.1971 | p=0.246 | rows=272 | events=96
MYCN-stratified event-rate: nonMYCN delta=0.077621 n=218 | MYCN-amp delta=0.242026 n=54
Component separation: antigenLoss delta=0.191176 | nkLow delta=0.352941 | combined delta=0.191176
Tier-1 package v3.0.4.2: TIER1_FULL_GPL5175_LOCAL_COX_TARGET | current=TIER0_DIRECTIONAL_SELECTED_MATRIX_ACTIVE | noStrictnessIncrease=true
GPL5175 annotation: FULL_GPL5175_ANNOTATION_REQUIRED_BEFORE_TIER1_PASS | fullPass=false
Verified-local Cox package: VERIFIED_LOCAL_COX_RERUN_CANDIDATE_RELAXED_MODE | label=SECONDARY_AXIS_VERIFIED_LOCAL_COX_RERUN_TARGET_APM_FALLBACK_SIGNAL_MECHANISM_NOT_D
Reproducibility receipt: sha256-10db8cba673903013b41322b887fd71322856ec207e4f1a7e7d8d360e180c006 | replay=REPLAY_PACKAGE_SPEC_READY_CURRENT_RUN_EMBEDDED_JSON_ONLY
Method claim: The app parses cancer hypotheses, maps them to expression-survival axes, runs relaxed first-pass event-rate/Cox screens, ranks directional signals, and packages reproducibility receipts while blocking official proof/kill claims before Tier-2.

Biology claim: This axis is interpreted only as a local directional result in the current selected-matrix screen.
Evidence readiness gate: BLOCKED_BY_MISSING_EVIDENCE | missing=official_or_declared_probe_platform_mapping,probe_to_gene_mapping_receipt,independent_replication_o
Evidence readiness receipts: platform/probe=PARTIAL_MAPPING_REVIEW_ONLY | outcomeResolved=true | covariateResolved=true | phenotypeComparator=SURVIVAL_OUTCOME_MODEL_NO_PHENOTYPE_COMPARATOR_REQUIRED_OR_NOT_DEC | replicationComparator=REPLICATION_COMPARATOR_NOT_DECLARED | confounder=CONFOUNDER_ADJUSTED_MODEL_EXECUTED_REVIEW_ONLY | mycnCoxExecuted=true | fragility=RAW_APM_SIGNAL_WEAKENED_AFTER_MYCN_ADJUSTMENT_MECHANISM_NOT_ATTRIBUTABLE
Axis continuation: AXIS_FOLLOW_UP_REQUIRED_NOT_NEW_DISCOVERY | next=Do not stop at promising. Run axis transfer, direction-stability, known-driver and bulk-attribution checks using the supplied axis only.
Operational readiness: BLOCKED_FROM_PROMOTION_BY_READINESS_GAPS — Resolve first: official_or_declared_probe_platform_mapping, probe_to_gene_mapping_receipt, independent_replication_or_cohort_transfer_check
Final hypothesis state v3.0.6.0: APM_FALLBACK_SIGNAL_MECHANISM_NOT_ATTRIBUTABLE | reason=APM-low direction may be tested locally, but CK2/CSNK2A1 mechanism is not directly attributable without mechanism-specific probes or perturbation context.
Decisive adjudication v3.0.6.5: verdict=KILLED_CK2_MECHANISM_ATTRIBUTION_APM_LOCAL_DIRECTION_SURVIVES | claimKilled=true | localSignalSurvives=true | notTestable=false | action=Stop mechanism claims; keep only the local APM direction for replication or full mapping.
Investment verdict v3.0.6.6: verdict=KILL_CK2_MECHANISM_DO_NOT_INVEST | killedClaim=CK2_CSNK2A1_directly_drives_APM_loss_and_worse_PFS_in_this_dataset | killCriteria=4 | investCriteria=1 | next=Do not invest in CK2. If continuing, invest only in mechanism-agnostic APM data, not CK2 experiments.
FINAL INVESTMENT DECISION v3.0.6.7: DO_NOT_INVEST_CLAIM_KILLED | claim=claim killed as written for current data | killedClaim=CK2_CSNK2A1_directly_drives_APM_loss_and_worse_PFS_in_this_dataset | criteria=kill:4/invest:1 | next=Do not invest in CK2. If continuing, invest only in mechanism-agnostic APM data, not CK2 experiments. | probabilityReported=false
PREPRINT READINESS v3.0.6.8: PREPRINT_READY_CLAIM_KILLED_AS_WRITTEN | title=Hypothesis Stress-Test & Investment Decision Report v3.0.6.8 | killedAsWritten=true | residualDataFollowUp=false | wetLabAllowed=false | probabilityReported=false
Publication data fuel v3.0.6.9: route=CK2_APM_FALLBACK | state=CK2_MECHANISM_REOPEN_GATES_INJECTED_CLAIM_REMAINS_KILLED | syntheticData=false | fakeReplication=false | defense=CK2 mechanism remains killed because external CK2/APM plausibility requires perturbation/protein/IFN context that current static bulk GSE85047 does not execute. | APM multi-cohort fuel v3.0.6.9.6: state=APM_MULTI_COHORT_DATA_FUEL_INJECTED_VALIDATION_TARGETS_ONLY | syntheticData=false | fakeReplication=false | targets=medulloblastoma, Wilms tumor/nephroblastoma, Ewing sarcoma | boundary=data-fuel only, not replication.
Pediatric cross-tumor scope v3.0.6.9.6: current=NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE | observed=neuroblastoma | crossTumorPromotionAllowed=false | wetLabPromotionAllowed=false
Pediatric expansion targets: medulloblastoma,Wilms tumor / nephroblastoma,Ewing sarcoma
Pediatric scope next action: After neuroblastoma, run the same stress-test template on medulloblastoma, Wilms tumor/nephroblastoma, and Ewing sarcoma; keep each as KILLED / NULL-LIKE / SURVIVING / NOT_MEASURABLE with separate receipts before any cross-tumor claim.

Pediatric scope boundary: Neuroblastoma-only evidence remains single-cohort and disease-scoped. Cross-pediatric-tumor credibility requires independent human datasets from at least three additional pediatric tumor families before any generalization, Tier-2 promotion, or wet-lab language.
Preprint boundary: This report supports a methods/resource-allocation claim only: supplied computational hypotheses are triaged as killed-as-written, watchlist-only, or data-only follow-up. It does not prove or disprove biology, does not recommend treatment, and does not establish clinical actionability.
Allowed manuscript sentence: The supplied claim is killed as written for the current data; the result should redirect resources away from the claimed mechanism rather than imply biological impossibility.
Direction: HIGH_SCORE_WORSE_PFS_EVENT_RATE | caveat=Relaxed first-pass screen only. Shows trial event-rate, MYCN-stratified and in-app Cox-review numbers for debugging. Not proof, not official support/kill, not clinical, not publication-grade, and not Tier-2 computational evidence.
Missing gates: official CSNK2A1 full GPL5175 mapping,IFN context not measurable,silmitasertib perturbation not testable,MHC-I protein abundance not available,full GPL5175 audit,source certification,independent Cox-capable replication,Tier-2 not reached,full GPL5175 mapping required before official support/kill,independent replication required before computational evidence language
Claim guard: BLOCKED_COMPUTATIONAL_CLAIM_REQUIRES_TIER2 | tier=TIER0_SLICE_REPRODUCIBLE_REVIEW_ONLY | officialPromotionAllowed=false | experimentalPromotionAllowed=false | experimentalPromotionForKilledClaimAllowed=false | dataFollowUpAllowed=false | computationalReplicationAllowed=false | wetLabAllowed=false | officialClaimAllowed=false
Tier-0 proves replayability, not source truth.
Tier-1 proves source extraction, not biological truth.
Tier-2 is required before any computational evidence claim.